Personal NAS & Private Cloud

Home Lab Project Documentation

Zero Trust | Self-Hosted | Cross-Platform

Field	Details
Purpose	Self-hosted personal cloud — replace paid services
Devices Supported	Windows Desktop, MacBook, iPhone
Storage	2x 1TB HDD (NTFS)
Remote Access	Tailscale VPN (WireGuard-based)
File Sharing	SMB / Windows Network Share
Security Model	Zero Trust + DAC ACL + MFA + Device Approval
Open Ports	None — all traffic via encrypted VPN tunnel
Subscription Cost	Free

Phase 1 — Drive Setup & Folder Structure

1.1 Drive Formatting

The NAS storage drives were formatted using Windows Disk Management to prepare them for NTFS-based network sharing.

Setting	Value
File System	NTFS
Allocation Unit Size	Default
Volume Label	CloudStorage
Format Type	Quick Format

1.2 Folder Structure

The following folder hierarchy was created on the CloudStorage volume to organize personal and professional data:

```
CloudStorage/
├── Documents/
├── Notes/
├── Videos/
├── Photos/
├── Projects/
├── Backups/ └── Shared/
```

Phase 2 — Network Sharing (SMB)

2.1 SMB Share Configuration

Windows built-in SMB (Server Message Block) was used to expose the CloudStorage volume as a network share. Configuration was done via the Drive's Properties → Sharing tab.

Setting	Value
Share Name	CloudStorage
Protocol	SMB (Server Message Block)
Access Scope	Tailscale network only (not exposed publicly)
Permissions	Per-user — not open to all

2.2 Windows Network Discovery

Configured in Control Panel → Network and Sharing Center:

- Network Discovery: Enabled
- File and Printer Sharing: Enabled
- Public sharing: Restricted — locked down at network level via Tailscale ACL

> ⚠ SMB is only accessible through the Tailscale VPN tunnel. It is NOT exposed on the public internet or the local LAN without VPN authentication.

Phase 3 — Tailscale VPN Overlay (Zero Trust)

3.1 Why Tailscale?

Traditional approaches to remote NAS access (port forwarding, DDNS, open SMB ports) expose your home network to the internet. Tailscale solves this by creating a private encrypted mesh network across all your devices — no open ports required.

> ℹ Tailscale uses the WireGuard protocol under the hood — one of the most modern, audited, and performant VPN protocols available.

3.2 Setup Steps

1. Installed Tailscale on Windows Desktop (NAS host), MacBook, and iPhone
2. All three devices enrolled into the same Tailscale account (tailnet)
3. Windows host configured to Never Sleep (required for always-on access)
4. NAS accessed via Tailscale-assigned IP address [REDACTED — 100.x.x.x range]

3.3 Network Architecture

```
[MacBook]  ──┐
             ├──▶ Tailscale Encrypted Mesh ──▶ [Windows NAS Host]
[iPhone]   ──┘      (WireGuard Protocol)              │
                                                  [2x 1TB HDD]
                                                 CloudStorage Share
 No open ports. No port forwarding. No public IP exposure.
```

Phase 4 — Security Policies

4.1 Tailscale Admin Portal Policies

Policy	Configuration
Admin Session Timeout	Auto-logout after inactivity period [duration redacted]
Device Approval Mode	MANUAL — all new devices require admin approval before access
Key Expiry	Devices must re-authenticate after set number of days [value redacted]
Tailscale SSH	Enabled — encrypted, no open SSH ports required

4.2 ACL Policy (JSON / DAC Model)

Access between devices is governed by a custom ACL (Access Control List) in the Tailscale admin portal. The model follows Discretionary Access Control (DAC) — the owner defines who can talk to what, and on which ports.

```
{
  "acls": [
    {
      "action": "accept",
      "src": ["tag:client"],
      "dst": ["tag:nas:445"]
    }
  ],
  "tagOwners": {
    "tag:nas": ["autogroup:owner"],
    "tag:client": ["autogroup:owner"]
  }
}
 // NOTE: Actual node names, IPs, and auth keys are REDACTED from this
documentation.
```

4.3 Tailscale SSH

Instead of running a traditional SSH server with an open port, Tailscale SSH was enabled. This provides:

- End-to-end encrypted SSH sessions
- Authentication tied to Tailscale identity — no SSH key management required
- No port 22 exposed on the machine
- Access logs and audit trail in the Tailscale admin console

4.4 Windows Firewall Rules

Custom firewall rules were configured to ensure SMB is completely inaccessible from the public internet or untrusted networks:

- SMB (port 445) blocked on all external/public interfaces
- SMB accessible only on the Tailscale virtual network interface
- Inbound and outbound rules reviewed and tightened

4.5 Security Hardening Summary

Security Measure	Status
Strong Windows Account Password	✅ Configured
SMB on External Interfaces	✅ Disabled

Multi-Factor Authentication (MFA)	✅ Enabled on Tailscale account
Device Authorization (Manual Approval)	✅ Enabled
Admin Session Auto-Logout	✅ Configured
Key/Session Re-Authentication Expiry	✅ Configured
Tailscale SSH (no open ports)	✅ Enabled
Open Router Ports (Port Forwarding)	✅ None — zero exposure
Public IP Exposure	✅ None

Security Considerations & Disclosures

⚠ This document has been sanitized for public sharing. All specific IPs, hostnames, Tailscale node names, auth keys, account identifiers, and exact policy values have been redacted or replaced with placeholders.

The following categories of information are intentionally excluded from this documentation:

- Tailscale node IPs and hostnames
- Tailscale auth keys or pre-authentication keys
- Windows account names or usernames
- Exact ACL policy rules beyond template examples
- Key expiry durations and session timeout values
- Device names as registered in the Tailscale admin console
- Physical network topology details

ⓘ This project is designed for personal/home lab use. For production or enterprise deployment, additional hardening such as log aggregation, intrusion detection, and formal incident response procedures would be required.

Key Concepts Demonstrated

Technologies & Concepts

Concept	Implementation
Self-Hosted Cloud Storage	Windows SMB network share over NTFS
Zero Trust Networking	Tailscale (WireGuard) — no implicit trust based on network location
Discretionary Access Control (DAC)	Tailscale ACL JSON policy — owner-defined access rules
Defense in Depth	Multiple overlapping layers: VPN + Firewall + MFA + Device Approval + SSH
Least Privilege	Devices only communicate on ports required for their function
Cross-Platform Access	Windows, macOS, and iOS all connected through a single VPN mesh
Always-On Server	Sleep prevention ensures NAS availability without enterprise UPS

References & Resources

The following resources were referenced during the design and implementation of this project:

- Tailscale Documentation: https://tailscale.com/kb
- Tailscale ACL Guide: https://tailscale.com/kb/1018/acls
- Tailscale SSH: https://tailscale.com/kb/1193/tailscale-ssh
- WireGuard Protocol: https://www.wireguard.com/
- Windows SMB Security Best Practices: https://learn.microsoft.com/en-us/windows-server/storage/file-server/smb-security
- NTFS Overview: https://learn.microsoft.com/en-us/windows-server/storage/file-system/ntfs-overview

Personal NAS — Home Lab Project

Documentation sanitized for public GitHub repository